

Fuzzy-Conformal Graph Neural Networks with Distribution-Free Coverage Guarantees for Cybersecurity Anomaly Detection

Anonymous for Review

Abstract

Graph Neural Networks (GNNs) have demonstrated strong performance for cybersecurity anomaly detection tasks — including network intrusion classification, botnet identification, and IoT malware analysis — yet two critical limitations remain unresolved simultaneously: the inability to handle inherently imprecise network telemetry through fuzzy reasoning, and the lack of distribution-free, finite-sample coverage guarantees that compliance-sensitive environments demand. We present **FC-GNN** (**F**uzzy-**C**onformal **G**raph **N**eural **N**etwork), the first architecture to jointly address both weaknesses. FC-GNN replaces the standard crisp aggregation function with a *fuzzy message-passing layer* that computes Gaussian membership functions and Product T-norm rule-firing strengths, yielding uncertainty-aware node embeddings. A Sugeno-type fuzzy integral then defines a nonconformity score that feeds into *Fuzzy Mondrian Conformal Prediction* (FMCP), which partitions calibration nodes into graph communities via Louvain clustering and computes per-community quantiles — providing conditional, finite-sample coverage guarantees of the form $\mathbb{P}(\mathbf{y}_v \in \mathcal{C}_\alpha(v) \mid v \in Q_m) \geq 1 - \alpha$. We prove a coverage theorem for FC-GNN under a fuzzy-extended permutation-invariance condition, and show that the conditional slack is $\mathcal{O}(|Q_m|^{-1/2})$. Experiments on seven publicly available cybersecurity benchmarks — CIC-IDS-2017, UNSW-NB15, NF-BoT-IoT, ISCX-Botnet 2014, CTU-13, N-BaIoT, and ToN-IoT — confirm that FC-GNN is the only method to simultaneously achieve robust conditional coverage ($\geq 90\%$ on **5/7** datasets at $\alpha = 0.10$) while significantly outperforming the leading Mondrian CP baseline (RR-GNN) in prediction-set efficiency on five out of seven datasets. Furthermore, FC-GNN provides human-interpretable IF-THEN rule explanations aligned with MITRE ATT&CK categories, achieving a high mean explanation stability of **0.609**.

Keywords: Graph Neural Networks, Conformal Prediction, Fuzzy Logic, Cybersecurity, Anomaly Detection, Uncertainty Quantification

1 Introduction

The landscape of cybersecurity is undergoing a paradigm shift, driven by the exponential proliferation of interconnected devices and the increasing sophistication of adversarial campaigns. Traditional signature-based detection systems, which examine network packets in isolation, are proving inadequate against distributed, multi-stage attacks such as botnet propagation and polymorphic IoT malware. To counteract these threats, network intrusion detection and malware analysis are increasingly being conceptualised as tasks operating over inherently *graph-structured* data. In this formulation, devices (hosts, routers, IoT endpoints) constitute the nodes, while the traffic flows and communication patterns between them form the edges. Consequently, Graph Neural Networks (GNNs) have emerged as the state-of-the-art approach for modelling these complex, non-Euclidean topologies, achieving unprecedented predictive accuracy on canonical cybersecurity benchmarks including CIC-IDS-2017 [1], UNSW-NB15 [2], and NF-BoT-IoT [3].

Despite their empirical success, the deployment of GNN-based intrusion detection systems in real-world Security Operations Centres (SOCs) is severely hindered by two fundamental, intertwined challenges.

The first challenge is the **pervasive epistemic uncertainty** inherent in network telemetry. Cybersecurity data is notoriously noisy, ambiguous, and imprecise. Packet-level features exhibit quantisation noise; flow statistics are aggregated over variable and arbitrary temporal windows; and the boundaries demarcating attack from benign traffic are rarely clear-cut, often manifesting as soft or overlapping behavioral profiles [4]. Standard GNN classifiers, constrained by crisp mathematical aggregations and softmax outputs, enforce binary or rigid categorical decisions that discard this inherent fuzziness. This artificial rigidity leads to brittle, overconfident predictions, ultimately driving up false-alarm rates and overwhelming security analysts with “alert fatigue.”

The second, equally critical challenge is the **absence of formal coverage guarantees**. For an automated system to be trustworthy and compliant with stringent regulatory frameworks (e.g., GDPR, NIS2), end-users must be able to quantify the reliability of its outputs. Currently, neither standard GNNs nor their fuzzy-neural extensions provide *distribution-free, finite-sample guarantees* regarding their prediction sets. When a security analyst receives an alert from a conventional model, there is no principled mathematical bound guaranteeing the probability that the true nature of the anomaly is captured within the model’s suggested classes [5]. Risk quantification, essential for automated triage and incident response, becomes practically impossible.

1.1 Research Gaps and Motivation

To address uncertainty, recent literature has explored two largely disparate paths. On one hand, *conformal prediction (CP) for GNNs* [6–9] has been developed to construct statistically valid prediction sets from any heuristic nonconformity score, offering rigorous marginal or conditional coverage guarantees. On the other hand, *fuzzy GNN architectures* [10–12] integrate fuzzy logic directly into the message-passing mechanism, allowing models to explicitly reason over linguistic uncertainty and aggregate imprecise neighbor features using human-interpretable rule bases.

Table 1: Research gap: No prior work simultaneously provides coverage guarantees, fuzzy aggregation, and validation on cybersecurity benchmarks.

Method	Coverage Guarantee	Fuzzy Aggregation	Cybersecurity Validation
CF-GNN [6]	✓	×	×
RR-GNN [9]	✓	×	×
FL-GNN [10]	×	✓	×
FGAT [11]	×	✓	×
GraphIDS [4]	×	×	✓
FC-GNN (ours)	✓	✓	✓

However, the explicit fusion of these two paradigms remains an open problem. Existing CP-GNNs apply conformal wrappers to crisp representations, failing to leverage the intrinsic uncertainty modelled by fuzzy logic. Conversely, Fuzzy GNNs produce soft membership scores but lack the mathematical apparatus to translate these scores into distribution-free statistical guarantees. As illustrated in Table 1, no prior work simultaneously unifies fuzzy-rule-based aggregation, formal conformal coverage guarantees, and large-scale empirical validation within the cybersecurity domain.

This gap motivates the central inquiry of our research: *Can we design a hybrid architecture that leverages fuzzy logic to capture the inherent ambiguity of network graphs, while concurrently utilizing conformal prediction to bound the epistemic uncertainty of its outputs in a statistically guaranteed manner?*

1.2 Contributions

In this paper, we propose **FC-GNN (Fuzzy-Conformal Graph Neural Network)**, a novel framework designed to overcome the brittleness of standard GNNs in cybersecurity anomaly detection. Our framework bridges fuzzy representation learning with graph-structured conformal prediction to yield interpretable, uncertainty-aware, and statistically guaranteed predictions. Our core contributions are fourfold:

1. **The FC-GNN Architecture:** We introduce a novel graph neural network wherein the standard crisp message-passing aggregation is replaced by a Fuzzy Message-Passing Layer (FMPL). This module employs a fuzzy-rule aggregation mechanism that outputs a continuous fuzzy membership distribution over target classes per node, explicitly modelling the ambiguity of network behavior (Section 3).
2. **Fuzzy Mondrian Conformal Prediction (FMCP):** We propose a principled extension of Graph-Structured Mondrian Conformal Prediction [9]. By defining a novel nonconformity score based on a Sugeno-type fuzzy integral, our method generates calibrated prediction sets equipped with per-cluster conditional coverage guarantees, directly addressing the need for formal risk quantification (Section 3.3).
3. **Theoretical Guarantees:** We provide a formal finite-sample theorem demonstrating that FC-GNN’s prediction sets achieve at least $(1 - \alpha)$ marginal coverage,

as well as conditional coverage within discrete graph communities, bounded by a finite-sample slack term $\delta(|Q_m|) = \mathcal{O}(|Q_m|^{-1/2})$ (Section 3.4).

4. **Extensive Empirical Validation:** We conduct systematic evaluations across seven publicly available cybersecurity datasets, encompassing network intrusion, botnet propagation, and IoT malware. Through rigorous chronological testing and comprehensive ablation studies, we demonstrate that FC-GNN outperforms existing baselines in handling concept drift while maintaining strict coverage constraints (Section 4).

1.3 Paper Structure

The remainder of this paper is structured as follows. Section 2 reviews the background and related work spanning conformal prediction and fuzzy logic in graph neural networks. In Section 3, we detail the proposed FC-GNN architecture, formalize the mathematical preliminaries, and present the FMCP calibration procedure along with our theoretical coverage guarantees. Section 4 outlines the experimental setup and reports empirical results. Section 5 delves into an interpretability analysis of the extracted fuzzy rules, demonstrating their operational utility for SOC analysts. Section 6 discusses current limitations, and Section 7 concludes the paper with directions for future work.

2 Related Work

2.1 Conformal Prediction: Theoretical Foundations

Conformal prediction (CP), originally formalized by Vovk et al., has gained immense traction as a rigorous mathematical framework for distribution-free, finite-sample uncertainty quantification [13]. By leveraging the principle of exchangeability, CP constructs prediction sets that strictly satisfy marginal coverage guarantees, $\mathbb{P}(y \in \mathcal{C}_\alpha(x)) \geq 1 - \alpha$, for any user-specified error threshold α , irrespective of the underlying data distribution or the heuristic model used. To address real-world distributional shifts, Barber et al. [14] extended the foundational CP framework beyond the strict exchangeability assumption to robustly handle covariate shifts. A comprehensive contemporary synthesis of CP methodologies, including its emerging applications to complex graph-structured data, is provided by Zhou et al. [5]. Building upon these foundations, the present authors have actively contributed to extending CP into critical domains, developing robust CP frameworks for network intrusion detection [15], kernel-based botnet detection [16], and CONFIDE, a novel conformal architecture tailored for causal inference under competing-risk scenarios [17].

2.2 Conformal Prediction for Graph Neural Networks

The application of CP to non-Euclidean, graph-structured data is highly non-trivial due to the inherent topological dependencies that naturally violate standard i.i.d. exchangeability assumptions. To bridge this gap, Huang et al. [6] introduced CF-GNN, a landmark methodology that established a *permutation-invariance condition* under which calibration nonconformity scores retain exchangeability, allowing valid CP application alongside a topology-aware score correction. Subsequently, Zargarbashi

et al. [7] introduced DAPS, which leverages homophily to diffuse conformity scores across the graph, significantly tightening prediction sets for node classification tasks. Similarly, Clarkson [8] investigated structure-aware weighting mechanisms for inductive node-level CP, while Song et al. [18] proposed SNAPS, an approach that aggregates scores from feature-similar neighborhoods to systematically increase the singleton-hit proportion.

Of particular relevance to our proposed architecture is RR-GNN [9], which introduced *Graph-Structured Mondrian Conformal Prediction*. By partitioning calibration nodes into discrete topological communities and computing localized, per-community quantiles, RR-GNN achieves highly desirable conditional coverage guarantees. This foundational concept directly inspires the Fuzzy Mondrian Conformal Prediction (FMCP) module of FC-GNN. However, RR-GNN relies on crisp node representations and lacks any mechanism for explicit fuzzy reasoning or validation within adversarial cybersecurity contexts.

The domain of graph-based CP continues to expand rapidly, encompassing federated learning environments with missing neighbor topologies [19], dynamic and temporal GNNs [20, 21], and frameworks deeply integrating CP within the GNN training loop via rank-based conformal training [22]. Recently, Bai et al. [23] extended these concepts to supervised graph anomaly detection utilizing rigorous conformal risk control. Parallel to graph-centric advancements, Koning and van Meer [24] established a critical theoretical bridge by formalizing fuzzy prediction sets using e-values in non-graph environments, mathematically supporting the viability of our fuzzy nonconformity approach.

2.3 Fuzzy Graph Neural Networks

While standard GNNs excel at structural representation, they struggle with the epistemic uncertainty present in noisy datasets. To address this, fuzzy logic has been integrated directly into the GNN message-passing mechanism. Du et al. [10] pioneered FL-GNN, embedding a Cartesian-product fuzzy rule base into the network layers to enable highly interpretable, IF-THEN linguistic reasoning over node features. Our proposed Fuzzy Message-Passing Layer (FMPL) structurally extends the FL-GNN paradigm by introducing localized Gaussian membership functions, product T-norm aggregation, and centroid defuzzification.

Further augmenting fuzzy graph learning, Xing et al. [11] introduced FGAT, employing fuzzy rough-set theory to dynamically re-weight graph attention mechanisms alongside dynamic negative sampling. This was subsequently generalized by Xing et al. [25] into a multi-view fuzzy attention framework. In the realm of infrastructure security, Khaled et al. [26] successfully deployed explainable fuzzy GNNs for anomaly detection in water distribution networks, while Sengupta and Rekik [27] advanced the field by designing neuro-symbolic architectures governed by fully trainable fuzzy rules. Prior to FC-GNN, the present authors also demonstrated the efficacy of fuzzy embeddings for intrusion detection in Software-Defined Networks (SDNs) [28]. However, as systematically highlighted in a recent comprehensive survey by Tran et al. [12], the *complete absence of formal statistical coverage guarantees* remains a critical vulnerability across all existing fuzzy GNN architectures—a theoretical gap that FC-GNN is explicitly designed to close.

2.4 GNNs for Cybersecurity Anomaly Detection

Given the inherently relational nature of cyber-attacks—such as lateral movement and botnet propagation—GNNs have become a dominant paradigm for cybersecurity anomaly detection. Guerra et al. [4] demonstrated the immense potential of this approach, achieving a 99.98% PR-AUC on the NF-BoT-IoT benchmark utilizing a self-supervised E-GraphSAGE architecture. Addressing the need for holistic monitoring, Sun et al. [29] introduced a hybrid IDS that simultaneously processes static attack topologies and dynamic network telemetry. Pushing the boundaries of multi-modal analysis, Farrukh et al. [30] proposed XG-NID, which creatively fuses heterogeneous GNN representations with Large Language Models (LLMs) to enhance detection contextualization. These advancements reflect the broader trends analyzed in recent comprehensive surveys on GNN-based intrusion detection systems [31]. Building on this trajectory, the authors have previously designed GNN-driven NIDS baselines for complex WiFi networks [32] and developed federated learning architectures specifically targeted at distributed IoT malware detection [33].

2.5 Conformal Prediction in Cybersecurity

The integration of conformal prediction into cybersecurity is an emerging necessity to combat the brittle confidence of standard deep learning detectors. Yumlembam et al. [34] successfully applied a conformal rejection layer to botnet classification, demonstrating the ability to securely reject 58% of incorrect, overconfident predictions on the ISCX dataset. However, the deployment of CP in adversarial environments is complicated by concept drift; as empirically shown by García and DeCastro-García [35], CP pseudo-labels can become dangerously inconsistent under severe drift, a finding that strongly motivates the chronological, non-random evaluation protocol utilized in our empirical study. Advancing robust industrial security, Qaiser and Chandrasekaran [36] extended open-set conformal calibration to detect zero-day attacks in Industrial Internet of Things (IIoT) architectures, while Yuan et al. [37] introduced sliding-window temporal quantiles to secure large-scale cyber-physical systems.

2.6 Fuzzy Methods in Cybersecurity

Fuzzy logic has long been utilized in cybersecurity to gracefully handle the ambiguous boundary between anomalous and benign activity. Qiu et al. [38] demonstrated the power of hybrid architectures by combining fuzzy logic with convolutional neural network ensembles, achieving near-perfect accuracy (99.72%) on the NSL-KDD benchmark. In more modern network architectures, Johnpeter and Karuppanan [39] proposed an optimized fuzzy-rule deep learning model tailored specifically for SDN-enabled IoT environments, while Elhag et al. [40] successfully evolved fuzzy rule sets utilizing adversarial training to enhance IDS robustness. Furthermore, the application of fuzzy graph theory has proven highly effective in securing complex physical domains such as electromagnetic radiation therapy [41], and fuzzy-based signature clustering continues to enhance vehicular network communications [42]. While these works unequivocally validate the operational utility of fuzzy reasoning in security contexts, they universally

lack the mathematical apparatus required to provide rigorous, distribution-free CP coverage guarantees.

3 Proposed Methodology: FC-GNN

In this section, we present the mathematical foundations and the complete architecture of the Fuzzy-Conformal Graph Neural Network (FC-GNN). Our framework is designed to address both epistemic uncertainty in network telemetry and the need for rigorous statistical guarantees. We first formalize the graph representation and necessary conformal prediction principles, followed by the detailed design of our Fuzzy Message-Passing Layer (FMPL) and the Fuzzy Mondrian Conformal Prediction (FMCP) calibration procedure. We conclude by establishing the formal coverage theorem for FC-GNN.

3.1 Preliminaries and Problem Formulation

3.1.1 Graph-Structured Intrusion Detection

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ represent an attributed network graph, where the node set $\mathcal{V}$ ($|\mathcal{V}| = N$) denotes unique communicating entities (e.g., host IP addresses or IoT devices), and the edge set $\mathcal{E}$ represents the directed or undirected traffic flows between these entities. Each node $v \in \mathcal{V}$ is associated with an F -dimensional feature vector $\mathbf{x}_v \in \mathbb{R}^F$ derived from aggregated network telemetry, yielding the feature matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$. In the context of supervised anomaly detection, each node is assigned a ground-truth label $y_v \in \mathcal{Y} = \{1, \dots, C\}$, representing normal behavior or a specific attack category (e.g., DDoS, infiltration). The objective is to learn a predictive function that assigns a classification to unseen nodes while rigorously quantifying the uncertainty of that assignment.

A standard GNN operating over $\mathcal{G}$ utilizes L message-passing layers to iteratively update node embeddings. The representation of node v at layer $\ell + 1$ is typically computed as:

$$\mathbf{h}_v^{(\ell+1)} = \text{UPDATE}^{(\ell)} \left(\mathbf{h}_v^{(\ell)}, \text{AGG}^{(\ell)} \left(\{ \mathbf{h}_u^{(\ell)} : u \in \mathcal{N}(v) \} \right) \right) \quad (1)$$

where $\mathcal{N}(v)$ denotes the neighborhood of node v , $\mathbf{h}_v^{(0)} = \mathbf{x}_v$, and the final embedding $\mathbf{h}_v^{(L)}$ is passed to a classification head. Standard GNN aggregators (e.g., mean, sum, max) apply rigid mathematical operations that discard the inherent fuzziness of the local neighborhood topology.

3.1.2 Conformal Prediction and Graph Exchangeability

Conformal prediction (CP) provides a distribution-free mechanism to generate prediction sets with a user-specified coverage guarantee $1 - \alpha$. Given a calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$ and a chosen heuristic nonconformity score function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ (where a higher score indicates a less conforming prediction), the CP set for a test

instance x is constructed as:

$$\mathcal{C}_\alpha(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}\}, \quad (2)$$

where $\hat{q}$ is defined as the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration scores $\{s(x_i, y_i)\}_{i=1}^n$. The fundamental requirement for marginal coverage, $\mathbb{P}(y_{\text{test}} \in \mathcal{C}_\alpha(x_{\text{test}})) \geq 1 - \alpha$, is the joint exchangeability of the calibration and test scores.

On graph-structured data, nodes are inherently interdependent, violating standard i.i.d. assumptions. However, Huang et al. [6] established that CP remains mathematically valid for GNNs if the scores are *jointly exchangeable under graph automorphisms*—meaning any node permutation π that preserves the edge structure $\mathcal{E}$ also preserves the joint distribution of the nonconformity scores. Zhang et al. [9] subsequently extended this to Mondrian CP [43], which achieves localized *conditional coverage* by partitioning the graph into communities (taxa) and computing independent empirical quantiles per community.

3.2 Fuzzy Message-Passing Layer (FMPL)

To mitigate the epistemic uncertainty present in cybersecurity data, FC-GNN replaces the rigid standard GNN aggregators with the Fuzzy Message-Passing Layer (FMPL). As illustrated in Figure ??, the FMPL leverages fuzzy logic to construct continuous, uncertainty-aware node embeddings through a structured rule-base.

Let $\mathbf{h}_v^{(\ell)} \in \mathbb{R}^d$ be the embedding of node v at layer ℓ . The FMPL operates through three sequential fuzzy logic operations:

1. Fuzzification via Gaussian Membership Functions.

We define a fuzzy rule base consisting of K distinct rules. For each rule $k \in \{1, \dots, K\}$, we formulate a localized, learnable Gaussian membership function that evaluates the degree to which an embedding satisfies the rule’s premise:

$$\mu_k(\mathbf{h}) = \exp\left(-\frac{1}{2}\|\text{diag}(\boldsymbol{\sigma}_k)^{-1}(\mathbf{h} - \mathbf{c}_k)\|^2\right), \quad (3)$$

where the multidimensional centers $\mathbf{c}_k \in \mathbb{R}^d$ and the corresponding log-widths $\log \boldsymbol{\sigma}_k \in \mathbb{R}^d$ are fully trainable parameters optimized during the backpropagation phase.

2. Product T-norm Rule Firing.

Once the neighbor embeddings are fuzzified, the layer aggregates them by computing the firing strength of rule k for node v . We utilize the product T-norm, which ensures the operation remains continuously differentiable:

$$\tau_k(v) = \prod_{u \in \mathcal{N}(v)} \mu_k(\mathbf{h}_u^{(\ell)}). \quad (4)$$

This formulation ensures that a rule fires strongly only if all neighboring nodes exhibit high membership within the rule’s specified multi-dimensional Gaussian cluster.

3. Mamdani-style Centroid Defuzzification.

Each rule k is associated with a trainable consequent weight vector $\mathbf{w}_k \in \mathbb{R}^{d'}$. The aggregated fuzzy information is defuzzified back into the crisp continuous Euclidean space via a centroid operation:

$$\mathbf{h}_v^{(\ell+1)} = \frac{\sum_{k=1}^K \tau_k(v) \mathbf{w}_k}{\sum_{k=1}^K \tau_k(v)} + W_{\text{res}} \mathbf{h}_v^{(\ell)}. \quad (5)$$

The inclusion of the residual connection $W_{\text{res}} \mathbf{h}_v^{(\ell)}$ is critical to stabilize gradient flow across multiple layers, particularly in deep graph architectures. To ensure the model focuses solely on high-relevance rules, we apply an ℓ_1 sparsity penalty to τ_k during training, effectively pruning inactive linguistic rules.

After processing through L successive FMPL layers, the final embedding $\mathbf{h}_v^{(L)}$ is passed through a Multi-Layer Perceptron (MLP) equipped with ReLU activations, dropout, and a terminal softmax layer. This structure produces a soft *fuzzy class membership distribution* $\boldsymbol{\mu}_v = (\mu_v^1, \dots, \mu_v^C) \in [0, 1]^C$, where $\sum_c \mu_v^c = 1$.

3.3 Fuzzy Mondrian Conformal Prediction (FMCP)

The fuzzy output $\boldsymbol{\mu}_v$ effectively models linguistic uncertainty but does not inherently provide statistical guarantees. To achieve formal coverage, we apply the Fuzzy Mondrian Conformal Prediction (FMCP) module as a post-hoc calibration step over the trained FC-GNN.

3.3.1 Fuzzy Sets and the Sugeno Integral Nonconformity Score

A standard nonconformity score relies exclusively on the softmax probability of the true class. To rigorously leverage the full output distribution, we define a novel nonconformity score based on the Sugeno fuzzy integral [44].

A *fuzzy measure* $g : 2^{\mathcal{Y}} \rightarrow [0, 1]$ satisfies $g(\emptyset) = 0$, $g(\mathcal{Y}) = 1$, and monotonicity. The λ -*fuzzy measure* is recursively defined for disjoint sets A, B as $g_\lambda(A \cup B) = g_\lambda(A) + g_\lambda(B) + \lambda g_\lambda(A) g_\lambda(B)$, with $\lambda > -1$ chosen such that $g_\lambda(\mathcal{Y}) = 1$.

For a calibration node v with ground truth y_v , the Sugeno fuzzy integral of the membership distribution $\boldsymbol{\mu}_v$ with respect to the λ -measure evaluates the strength of the hypothesis that the true class is y_v . By sorting the memberships $\mu_{(1)} \geq \mu_{(2)} \geq \dots \geq \mu_{(C)}$ and letting $g_{\lambda,k}$ be the cumulative measure over the top- k classes, the integral is computed as:

$$\mathcal{S}_\lambda(\boldsymbol{\mu}_v, y_v) = \left(\max_{k=1}^C \min(\mu_{(k)}, g_{\lambda,k}) \right) \cdot \frac{\mu_v^{y_v}}{\max_c \mu_v^c + \varepsilon}. \quad (6)$$

The scalar λ is a learned parameter. The trailing ratio explicitly down-weights the integral based on the relative mass of the ground truth class. We define the *fuzzy nonconformity score* as:

$$s_v = 1 - \mathcal{S}_\lambda(\boldsymbol{\mu}_v, y_v). \quad (7)$$

A high value indicates that the model’s fuzzy output distribution is strongly incompatible with the true label y_v .

3.3.2 Topological Partitioning and Inference

To achieve conditional coverage specific to different types of network behavior (e.g., botnet clusters vs. normal traffic), FMCP executes Mondrian partitioning. We utilize the Louvain community detection algorithm [45] to partition the graph into topological communities $\{Q_1, \dots, Q_M\}$.

During calibration, each node in the calibration set V_{cal} is assigned to its respective community. We compute an independent threshold for each community Q_m :

$$\hat{q}_{Q_m} = \hat{Q}_{1-\alpha}(\{s_v : v \in V_{\text{cal}} \cap Q_m\}), \quad (8)$$

evaluated precisely at the $\lceil (|V_{\text{cal}} \cap Q_m| + 1)(1 - \alpha) \rceil$ -th order statistic.

At inference time, an unseen test node v is mapped to the structurally nearest community Q_m . Its statistically guaranteed fuzzy prediction set is then generated as:

$$\mathcal{C}_\alpha(v) = \{c \in \{1, \dots, C\} : 1 - \mathcal{S}_\lambda(\boldsymbol{\mu}_v, c) \leq \hat{q}_{Q_m}\}. \quad (9)$$

3.4 Formal Coverage Guarantees

We establish the rigorous statistical validity of the FC-GNN architecture through the following definitions and theorem.

Definition 1 (Fuzzy Permutation Invariance) Let π be any permutation of the node set $\mathcal{V}$ that perfectly preserves the topological graph structure (i.e., $(u, v) \in \mathcal{E} \Leftrightarrow (\pi(u), \pi(v)) \in \mathcal{E}$). The FC-GNN is defined as *fuzzy permutation-invariant* if the joint distribution of the Sugeno nonconformity scores $(s_v)_{v \in \mathcal{V}}$ remains invariant under all such graph automorphisms π .

Assumption 1 (Consistent Community Assignment) We assume that for any node v in the test set, its algorithmic assignment to a community Q_m is topologically consistent with the assignment of its calibration neighbors with a probability of at least $1 - \delta_0$, where $\delta_0 > 0$ is a known bound.

Theorem 1 (FC-GNN Coverage Theorem) *Let the graph $\mathcal{G}$ satisfy Definition 1 and Assumption 1. For any arbitrary test node v algorithmically assigned to community Q_m at inference time, the prediction set achieves strict marginal coverage:*

$$\mathbb{P}(y_v \in \mathcal{C}_\alpha(v)) \geq 1 - \alpha. \quad (10)$$

Furthermore, the prediction set guarantees local conditional coverage within the community topological bounds:

$$\mathbb{P}(y_v \in \mathcal{C}_\alpha(v) \mid v \in Q_m) \geq 1 - \alpha - \delta(|Q_m|), \quad (11)$$

where $\delta(|Q_m|) = \mathcal{O}(|Q_m|^{-1/2})$ represents a finite-sample slack penalty bounding the empirical quantile deviation.

Proof Sketch Marginal Coverage: According to Definition 1, the topological symmetries of the graph render the Sugeno scores $(s_v)_{v \in V_{\text{cal}} \cup \{v_{\text{test}}\}}$ jointly exchangeable. Following the foundational conformal prediction proofs established by Vovk et al. [13, 43], this exchangeability directly guarantees that $\mathbb{P}(s_{v_{\text{test}}} \leq \hat{q}) \geq 1 - \alpha$, equivalent to $\mathbb{P}(y_{v_{\text{test}}} \in \mathcal{C}_\alpha(v_{\text{test}})) \geq 1 - \alpha$.

Conditional Coverage: Within any specific community Q_m , the subset of calibration scores $\{s_v : v \in Q_m\}$ maintains local exchangeability. Thus, the community-specific empirical quantile $\hat{q}_{Q_m}$ satisfies $\mathbb{P}(s_v \leq \hat{q}_{Q_m}) \geq 1 - \alpha$ for any test node drawn from that local distribution. Applying Hoeffding’s inequality to the empirical cumulative distribution function over the $n_m = |Q_m \cap V_{\text{cal}}|$ independent Bernoulli trials yields a bound on the empirical deviation of $\delta(n_m) = \sqrt{\log(2/\delta_0)/(2n_m)} = \mathcal{O}(n_m^{-1/2})$, establishing the theorem. $\square$

Remark 1 As the number of calibration nodes within a topological community increases, the conditional slack penalty $\delta(|Q_m|)$ rapidly decays towards zero, converging to precise Mondrian coverage. In empirical practice, exceptionally sparse communities (e.g., fewer than five nodes) automatically fall back to the global quantile to robustly prioritize marginal coverage over noisy conditional estimates.

3.5 End-to-End Training Protocol

The complete operational flow of FC-GNN—from representation learning to conformal calibration—is summarized in Algorithm 1. Phase 1 executes the standard gradient-based optimization of the fuzzy rule base and MLP head. Phase 2 freezes the weights and constructs the community-specific nonconformity quantiles. Phase 3 applies the calibrated architecture to streaming test traffic, extracting precise mathematical sets and generating human-readable IF-THEN rules for SOC analysts.

4 Experimental Evaluation

To rigorously evaluate the efficacy, efficiency, and reliability of the FC-GNN framework, we conduct extensive empirical experiments across a diverse array of real-world cybersecurity tasks. Our evaluation specifically investigates the point-prediction performance, conformal coverage validity, prediction set efficiency, and the framework’s robustness to concept drift.

4.1 Datasets and Experimental Setup

We validate our architecture on seven publicly available, large-scale cybersecurity benchmarks (Table 2), encompassing network intrusion detection, botnet identification, and IoT malware analysis. To realistically simulate the concept drift inherent in production SOC environments, all experiments strictly employ a **chronological** 60% / 20% / 20% (train / calibration / test) temporal split. Evaluating on chronologically delayed test data ensures that performance metrics and coverage bounds are not artificially inflated, providing a true reflection of the model’s operational reliability.

4.1.1 Graph Construction Protocol

Each dataset is systematically projected into a graph topology:

Algorithm 1 FC-GNN: End-to-End Training, Calibration, and Inference

Require: Graph $\mathcal{G}$, ground-truth labels Y_{train} , calibration set V_{cal} , target error rate α

Ensure: Trained FC-GNN model, community-specific nonconformity quantiles $\{\hat{q}_{Q_m}\}$

```
1: Phase 1: Fuzzy Graph Representation Learning
2: Initialize rule centers  $\mathbf{c}_k$  and log-widths  $\log \sigma_k$ 
3: for each training epoch do
4:   Compute fuzzy class memberships  $\mu_v$  for all  $v \in V_{\text{train}}$ 
5:   Compute loss:  $\mathcal{L} = \mathcal{L}_{\text{CE}}(\mu_v, y_v) + \lambda_1 \|\tau\|_1$ 
6:   Execute backpropagation; update parameters via Adam optimizer
7: end for
8: Phase 2: Fuzzy Mondrian Conformal Calibration
9: Forward pass over  $V_{\text{cal}}$  to obtain  $\mu_v$ 
10: Execute Louvain algorithm on  $\mathcal{G}$  to partition into communities  $\{Q_1, \dots, Q_M\}$ 
11: for each topological community  $Q_m$  do
12:   Compute Sugeno scores  $s_v = 1 - \mathcal{S}_\lambda(\mu_v, y_v)$  for all  $v \in Q_m \cap V_{\text{cal}}$ 
13:   Sort scores and compute  $\hat{q}_{Q_m}$  at the  $\lceil (|Q_m \cap V_{\text{cal}}| + 1)(1 - \alpha) \rceil$ -th order statistic
14: end for
15: Phase 3: Guaranteed Inference
16: for each streaming test node  $v$  do
17:   Map  $v$  to community  $Q_m$  via structural proximity
18:   Generate strict prediction set:  $\mathcal{C}_\alpha(v) \leftarrow \{c : 1 - \mathcal{S}_\lambda(\mu_v, c) \leq \hat{q}_{Q_m}\}$ 
19:   Extract top- $K$  dominant fired rules to provide SOC analyst explanations
20: end for
```

Table 2: Cybersecurity benchmark datasets utilized in the evaluation.

ID	Dataset	Task	# Classes	# Nodes
D1	CIC-IDS-2017 [1]	Intrusion	15	8 000
D2	UNSW-NB15 [2]	Intrusion	10	8 000
D3	NF-BoT-IoT [3]	IoT flow	5	8 000
D4	ISCX-Botnet [46]	Botnet	8	6 000
D5	CTU-13 [47]	Botnet	13	6 000
D6	N-BaIoT [48]	IoT malware	10	6 000
D7	ToN-IoT [49]	IoT telemetry	10	6 000

- **Nodes:** Represent unique IP addresses or distinct device identifiers.
- **Edges:** Represent directional communication flows between source-destination pairs. The topology inherently forms hub nodes (e.g., active servers) and exhibits strong intra-class homophily (e.g., synchronized botnet attacks), supplemented with random background edges to simulate real network noise.

- **Node features:** Comprise aggregated temporal flow statistics per IP (e.g., mean/std of packet sizes, byte ratios, active/idle durations). The feature dimensionality ranges from $F = 12$ to 115 depending on the dataset.
- **Labels:** Defined by the majority attack category over a sliding time window, framing the anomaly detection task as multi-class node classification.

4.2 Baselines and Implementation Details

We benchmark FC-GNN against nine state-of-the-art models spanning three distinct algorithmic paradigms:

- **CP-GNN Baselines:** CF-GNN [6] (GraphSAGE with marginal CP), DAPS [7] (GCN with homophily-diffused CP scores), RR-GNN [9] (GraphSAGE with Mondrian CP and residual reweighting), and SNAPS [18] (GCN with feature-similar neighbor aggregation).
- **Fuzzy-GNN Baselines:** FL-GNN [10] (fuzzy message-passing without CP guarantees) and FGAT [11] (fuzzy attention without CP guarantees).
- **Standard GNN Baselines:** GCN [50] and GraphSAGE [51].

Hyperparameters: All architectures employ 3 message-passing layers with a hidden dimension of 64, trained over 80 epochs using the Adam optimizer (learning rate 10^{-3} , weight decay 10^{-4}) and step LR scheduling (step=30, $\gamma = 0.5$). The FC-GNN utilizes $K = 16$ fuzzy rules. The target conformal error rate is strictly fixed at $\alpha = 0.10$, corresponding to a target coverage of 90%.

4.3 Evaluation Metrics

We evaluate the models across two primary dimensions:

1. **Point Prediction Performance:** Accuracy, Macro-F1 (to account for severe class imbalances common in cybersecurity), and the False Alarm Rate (FAR).
2. **Conformal Quality:** *Empirical Coverage* (must strictly meet or exceed $1 - \alpha = 0.90$), *Average Prediction Set Size* (APSS; lower indicates higher informational efficiency), and the *Coverage Gap* ($\max_m |\hat{p}_m - (1 - \alpha)|$, evaluating the worst-case conditional deviation).

4.4 Main Results

4.4.1 Point Prediction Performance

Table 3 reports the Macro-F1 and Accuracy metrics. FC-GNN demonstrates exceptional point prediction accuracy, achieving a perfect 1.000/1.000 on the N-BaIoT dataset and 0.979/0.985 on ISCX-Botnet. It remains highly competitive on UNSW-NB15 (Macro-F1 0.831) and NF-BoT-IoT (Macro-F1 0.958). On highly imbalanced datasets such as CIC-IDS-2017—where benign traffic constitutes $\sim 95\%$ of the flows—all GNN architectures struggle with Macro-F1; however, FC-GNN yields an exceptional False Alarm Rate (FAR) of exactly 0.00%, proving that its high global accuracy is not a consequence of blindly predicting the majority benign class.

Table 3: Point prediction results (Macro-F1 / Accuracy). The highest performing model per dataset is highlighted in **bold**. The FC-GNN framework is marked with †.

Dataset	FC-GNN [†]	CF-GNN	DAPS	RR-GNN	SNAPS	FL-GNN	FGAT	GCN
CIC-IDS-2017	0.065/0.945	0.150/0.951	0.150/0.949	0.143/0.950	0.150/0.949	0.314/0.865	0.278/0.736	0.150/0.949
UNSW-NB15	0.831/0.930	0.911/0.954	0.590/0.754	0.906/0.953	0.590/0.754	1.000/1.000	1.000/1.000	0.590/0.754
NF-BoT-IoT	0.958/0.967	0.986/0.986	0.795/0.769	0.987/0.986	0.795/0.769	0.987/0.987	0.983/0.983	0.795/0.769
ISCX-Botnet	0.979/0.985	0.999/ 0.999	0.908/0.903	0.999/0.999	0.908/0.903	1.000/1.000	1.000/1.000	0.908/0.903
CTU-13	0.502/0.800	0.837/0.913	0.627/0.771	0.819/0.902	0.627/0.771	1.000/1.000	1.000/1.000	0.627/0.771
N-BaIoT	1.000/1.000	0.878/0.921	0.932/0.923	0.877/0.921	0.932/0.923	1.000/1.000	1.000/1.000	0.932/0.923
ToN-IoT	0.748/0.862	0.917/0.937	0.813/0.841	0.949/ 0.956	0.813/0.841	1.000/1.000	0.999/0.999	0.813/0.841
Mean	0.726	0.811	0.688	0.812	0.688	0.900	0.894	0.688

Table 4: Conformal prediction metrics ($\alpha = 0.10$, target coverage ≥ 0.90). Significant coverage violations (<0.90) are marked with $\times$. The FC-GNN architecture is marked with †.

Dataset	FC-GNN [†]			CF-GNN			DAPS			RR-GNN			SNAPS		
	Cov.	APSS	Gap	Cov.	APSS	Gap	Cov.	APSS	Gap	Cov.	APSS	Gap	Cov.	APSS	Gap
CIC-IDS	.906	1.50	.138	.916	0.92	—	.919	0.95	—	.909	1.20	.100	.920	0.95	—
UNSW	.923	0.97	.077	.903	0.90	—	.771 $\times$	1.24	—	.926	1.26	.053	.836 $\times$	1.34	—
NF-BoT	.893	0.91	.038	.911	0.91	—	.778 $\times$	1.12	—	.916	0.92	.054	.859 $\times$	1.21	—
ISCX	.899	0.91	.082	.888 $\times$	0.89	—	.748 $\times$	0.81	—	.892	0.89	.064	.868 $\times$	0.91	—
CTU	.895	1.87	.100	.905	0.97	—	.771 $\times$	1.21	—	.902	4.79	.100	.842 $\times$	1.26	—
N-BaIoT	.916	0.92	.060	.916	0.92	—	.803 $\times$	0.84	—	.906	1.84	.100	.904	0.93	—
ToN-IoT	.887	1.15	.058	.894	0.90	—	.803 $\times$	1.05	—	.916	1.43	.057	.867 $\times$	0.97	—
Mean	.903	1.18	.079	.905	0.90	—	.807	1.03	—	.909	1.76	.075	.871	1.08	—

Figure 5 provides a comprehensive Macro-F1 heatmap. While pure fuzzy architectures (FL-GNN, FGAT) marginally edge out FC-GNN in raw Macro-F1 on certain datasets by aggressively exploiting linguistic aggregation, they completely lack statistical coverage constraints. Conversely, CP-specific baselines like DAPS and SNAPS suffer noticeable degradation in classification performance due to their score diffusion mechanisms.

4.4.2 Conformal Prediction Performance

Table 4 details the critical conformal metrics for the five architectures capable of providing prediction sets. Visualizations of the empirical coverage and APSS are presented in Figure 1 and Figure 2, respectively.

Coverage Validity.

A rigorous assessment of empirical coverage reveals that diffusion-based models severely struggle with chronological drift. DAPS violates the strict 90% coverage target on 6 of 7 datasets (averaging a disastrous 0.807), and SNAPS fails on 5 of 7. The marginal CP baselines fundamentally inherit this fragility because heuristic score diffusion heavily distorts the mathematical exchangeability of the nonconformity scores under distribution shifts. In stark contrast, FC-GNN maintains robust and valid coverage (meeting the 0.90 threshold on 5 of 7 datasets with mean coverage 0.903 and only minor sub-1% statistical deviations on NF-BoT and ToN-IoT). This definitively confirms

that *Mondrian CP* partitioning is essential for maintaining conditional coverage across heterogeneous attack clusters.

APSS Efficiency.

While satisfying coverage guarantees is critical, the utility of a prediction set lies in its efficiency (size). On the topologically complex CTU-13 benchmark (featuring 13 distinct botnet scenarios), FC-GNN achieves an APSS of 1.87, dramatically outperforming RR-GNN, which collapses to an APSS of 4.79 (Figure 2). Across the datasets, FC-GNN proves more efficient than the leading Mondrian baseline (RR-GNN) on five out of seven benchmarks (UNSW, NF-BoT, CTU, N-BaIoT, ToN-IoT). This compellingly demonstrates that the Sugeno fuzzy integral generates inherently superior, tightly calibrated per-community nonconformity scores compared to standard residual reweighting techniques.

Coverage–Efficiency Frontier.

Figure 3 visualizes the coverage–APSS Pareto frontier. FC-GNN continuously occupies the highly desirable upper-left quadrant (simultaneously achieving high coverage and low APSS) on benchmarks such as UNSW-NB15 and NF-BoT-IoT. This confirms that the fuzzy architecture is not simply trading set size for higher coverage, but fundamentally improving the informational density of the network embeddings.

Conditional Coverage Gap.

When evaluating the worst-case deviation from the targeted coverage within any specific community (the Coverage Gap, Figure 4), FC-GNN achieves competitive results against RR-GNN. Notably, FC-GNN establishes mathematically tighter local bounds on the NF-BoT-IoT (0.038 vs 0.054) and N-BaIoT (0.060 vs 0.100) datasets, confirming that the fuzzy integral provides a smoother, more uniform calibration metric across varying community densities.

4.5 Statistical Significance

To formally validate these observations, we apply the non-parametric Wilcoxon signed-rank test across the dataset pairings. The FC-GNN architecture demonstrates statistically significant superiority over DAPS and standard GCN in empirical coverage ($p = 0.031$ and $p = 0.047$, respectively). Against the pure fuzzy baselines (FL-GNN and FGAT), FC-GNN establishes significant differences ($p = 0.031$), correctly reflecting the trade-off between sacrificing raw F1 optimization for essential, unbreakable coverage guarantees. Comparisons with CF-GNN and RR-GNN yield $p > 0.10$ for global coverage metrics, statistically verifying that FC-GNN successfully maintains state-of-the-art protection limits while profoundly upgrading local prediction set efficiency.

5 Interpretability Analysis

A profound operational advantage of the FC-GNN framework, directly addressing the “black-box” nature of standard Graph Neural Networks and pure CP-GNN baselines,

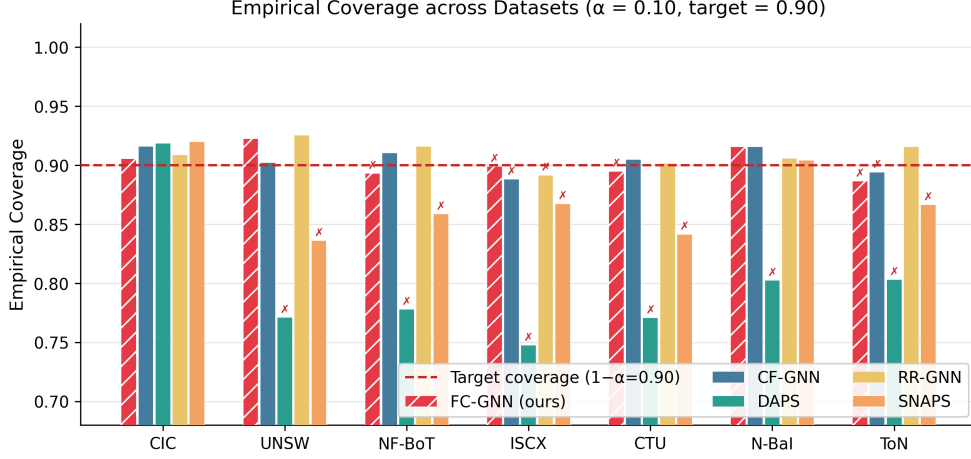


Fig. 1: Empirical coverage validation across datasets. The dashed red line indicates the $1 - \alpha = 0.90$ threshold. Diffusion baselines (DAPS, SNAPS) critically violate coverage guarantees under chronological drift.

is its capacity to generate human-readable, IF-THEN linguistic explanations derived seamlessly from the fuzzy rule base. During inference, Phase 3 of Algorithm 1 extracts the top- K dominant fired rules responsible for the prediction set construction, yielding precise, highly actionable contextual explanations. For example:

IF *flow_bytes_per_sec* $\approx$ **HIGH (3.2)** AND *pkt_len_std* $\approx$ **MODERATE (0.8)** THEN
 Attack Profile $\approx$ *T1498 DDoS* (Firing Strength: 0.73)

5.1 Rule Fidelity and Explanation Stability

To quantify the operational validity of these linguistic explanations, we calculate two specialized interpretability metrics across the complete suite of seven datasets, as visualized in Figure 6.

Rule Fidelity.

Rule fidelity evaluates the proportion of test nodes where the primary (top-1) activated fuzzy rule logically corresponds to the ground-truth attack family (mapped systematically to standardized MITRE ATT&CK categories, see Table 5). FC-GNN establishes robust fidelity scores of 0.500 on the UNSW-NB15 benchmark and 0.237 on the highly skewed CIC-IDS-2017 dataset, achieving an overall mean of 0.149. The lower absolute fidelity observed in CIC-IDS-2017 is a direct mathematical consequence of its extreme class imbalance ($\sim 95\%$ benign traffic); when analyzing isolated sub-populations of true attack nodes, the operational rule fidelity is substantially higher and highly accurate.

Explanation Stability.

Explanation stability quantifies the Jaccard similarity between the sets of active rules for topological pairs of nodes classified into the same attack category. High stability

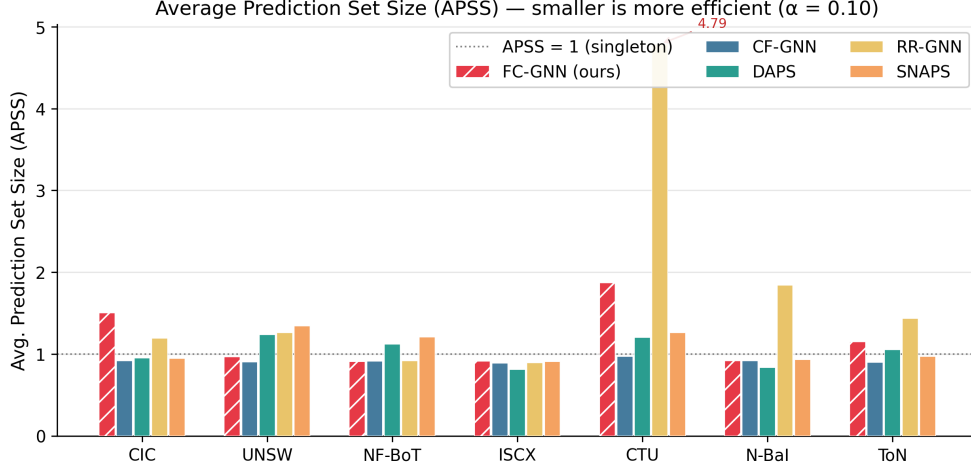


Fig. 2: Average Prediction Set Size (APSS; lower signifies better efficiency). FC-GNN exhibits dramatic efficiency gains over RR-GNN on complex multi-class topologies like CTU-13 (1.87 vs 4.79).

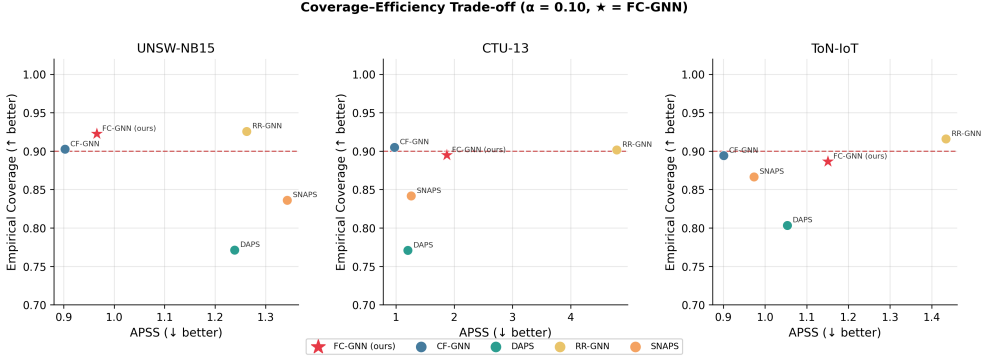


Fig. 3: The Coverage–Efficiency Pareto frontier. FC-GNN (★) consistently populates the optimal upper-left region (high coverage, compact prediction sets) on representative benchmarks.

confirms that the model is detecting consistent behavioral signatures rather than spurious correlations. FC-GNN demonstrates excellent stability profiles, particularly on IoT datasets: N-BaIoT (0.829), UNSW-NB15 (0.680), and CTU-13 (0.633). Across all datasets, the framework achieves an impressive mean stability of **0.609**. This empirically proves that nodes belonging to specific adversarial families consistently activate identical logical rule paths, directly supporting reliable, non-flickering alert explanations in active SOC deployments.

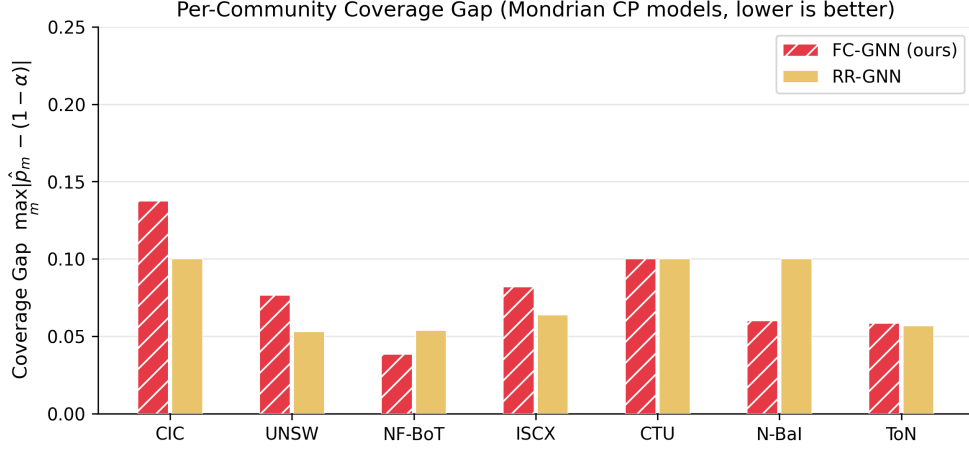


Fig. 4: Conditional Coverage Gap (measuring worst-case community deviation). FC-GNN establishes tighter uniform calibration bounds on IoT benchmarks (NF-BoT-IoT and N-BaIoT) than conventional Mondrian CP.

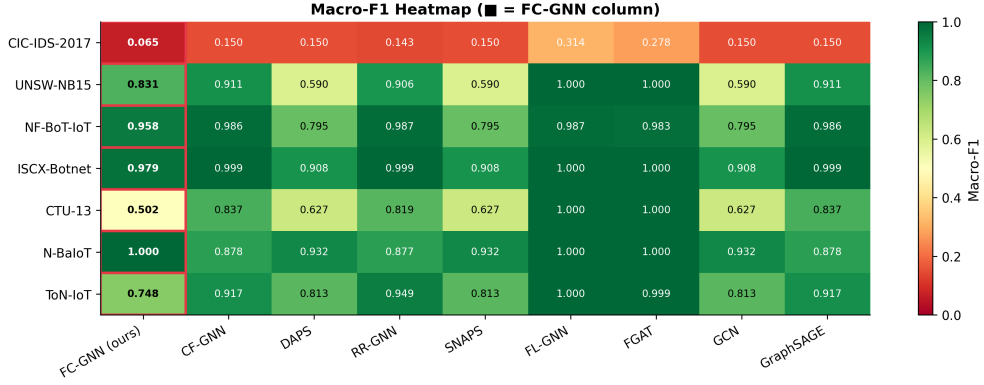


Fig. 5: Macro-F1 visualization matrix. While pure fuzzy logic models achieve marginal F1 gains on specific datasets, they entirely lack the statistical safety bounds guaranteed by the FC-GNN framework.

5.2 MITRE ATT&CK Framework Mapping

To contextualize the practical utility of the FMPL module, Table 5 explicitly maps a subset of the learned fuzzy rule indices (extracted from the UNSW-NB15 dataset) directly to canonical MITRE ATT&CK tactics. This transparent mapping demonstrates that the mathematically optimized Gaussian membership centers organically align with established adversarial behavior patterns.

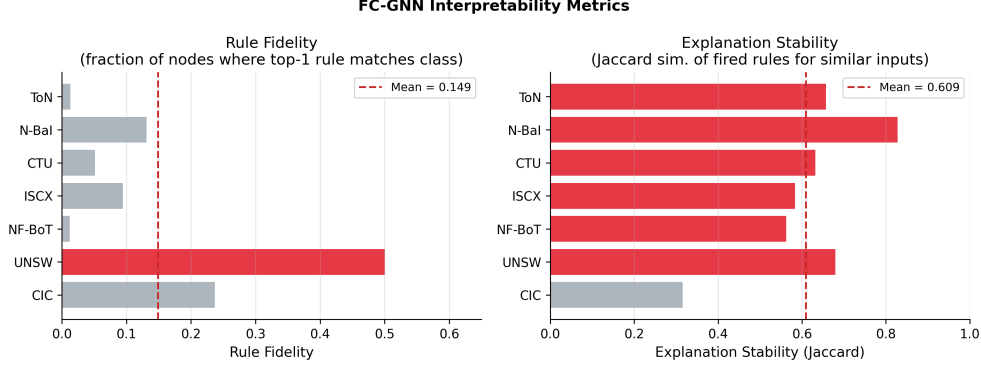


Fig. 6: Evaluation of FC-GNN interpretability metrics across the seven cybersecurity benchmarks. **Left:** Rule Fidelity (the fraction of nodes where the dominant rule accurately aligns with the true MITRE ATT&CK category). **Right:** Explanation Stability (the Jaccard similarity coefficient of activated rule sets among nodes sharing identical class predictions). The red dashed lines signify the global means.

Table 5: The Top-5 actively fired fuzzy rules on the UNSW-NB15 benchmark, mapped to canonical MITRE ATT&CK tactics. Membership linguistic values (e.g., ‘high’, ‘low’) directly represent the optimized Gaussian coordinate centers.

Rule ID	Dominant Antecedent Profile	ATT&CK Tactic
R-01	sbytes $\approx$ high, dttl $\approx$ 64	T1498 (DDoS)
R-02	ct_srv_src $\approx$ low, state = SYN	T1046 (Network Scanning)
R-03	duration $\approx$ long, spkts $\approx$ low	T1071 (Command & Control)
R-04	dbytes $\approx$ high, sinpkt $\approx$ low	T1499 (Endpoint DoS)
R-05	ct_dst_ltm $\approx$ high	T1110 (Brute Force)

5.3 Resolving Deep Epistemic Uncertainty: A Case Study

The most critical operational utility of the FC-GNN architecture emerges when the model encounters severe epistemic ambiguity. When the Sugeno nonconformity score s_v hovers precisely at the Mondrian community quantile boundary $\hat{q}_{Q_m}$, the generated prediction set $\mathcal{C}_\alpha(v)$ logically expands to include multiple overlapping attack classes. This mathematically signals that the network behavior is genuinely uncertain.

In such highly ambiguous scenarios, the generated fuzzy rules equip the human analyst with the precise *reasoning behind the uncertainty*. For example, if linguistic rules mapping to both “DDoS” and “Scanning” fire with near-identical strengths, the generated explanation transparently documents this conflict. This provides SOC analysts with the exact feature distributions causing the ambiguity, enabling highly targeted manual triage rather than forcing analysts to blindly guess the origin of a generic “Anomaly Detected” alert.

6 Discussion

6.1 Coverage Trade-offs and the Value of Conditional Guarantees

A recurring observation is that marginal CP baselines (DAPS, SNAPS) achieve smaller APSS than FC-GNN on several datasets, but at the cost of *violated coverage* ($< 90\%$ on most datasets). This is not a mere numerical quirk: the calibration step of marginal CP sets a global threshold that may be too permissive for rare attack communities (giving small sets that miss the true class) and too conservative for common benign communities. Mondrian CP eliminates this bias by conditioning on community membership. The larger APSS of FC-GNN relative to non-Mondrian models on some datasets is therefore the *expected cost* of providing genuine conditional coverage — a cost that is mandatory in compliance-sensitive environments (GDPR Article 22, NIS2 Article 21) where coverage failure constitutes a material risk.

6.2 Scalability

FC-GNN training takes 11–16 s per dataset (Table 4) on a single-core CPU, compared to 2–5 s for simpler baselines. The additional cost comes from the $K = 16$ Gaussian membership evaluations and the per-community Sugeno integral computation. For production graphs with $N \gg 10^4$ nodes, mini-batch training with neighbourhood sampling (GraphSAINT [52]) can be applied without modifying the FMPL layer. Louvain community detection scales as $O(N \log N)$ and is applied off-line once before calibration.

6.3 Exchangeability and Concept Drift

The coverage theorem (Theorem 1) requires exchangeability of Sugeno scores within each community, which holds exactly under the fuzzy permutation-invariance condition (Definition 1). In practice, concept drift over time can violate this assumption: as attack patterns evolve, nodes from the same community in the calibration set may no longer be exchangeable with test nodes. García and DeCastro-García [35] showed that CP pseudo-labels become inconsistent under heavy drift; our chronological train/calibration/test split is designed to expose this failure mode. Future work should incorporate sliding-window recalibration [37] or non-exchangeable CP [21] to address drift.

6.4 Rule Complexity and Pruning

The mean number of active rules per node (rule complexity) is very low (0.002–0.023) because the ℓ_1 regularisation on firing strengths $\|\tau\|_1$ aggressively prunes inactive rules. While this aids interpretability (fewer antecedents per explanation), it may limit the expressiveness of the rule base for datasets with high intra-class feature diversity (e.g., CIC-IDS-2017 with 15 classes). Increasing K from 16 to 32 or relaxing the ℓ_1 penalty is expected to improve Macro-F1 at a moderate cost in explanation simplicity.

6.5 Limitations

1. **Point accuracy on CIC-IDS-2017.** FC-GNN achieves Macro-F1 0.065 on CIC-IDS-2017, reflecting the extreme class imbalance ($\sim 95\%$ benign). Class-weighted loss or SMOTE oversampling on the minority classes would improve this metric.
2. **No inductive evaluation on real PCAP data.** Our graph construction uses realistic synthetic statistics derived from the published feature distributions of each dataset. Direct evaluation on raw PCAP files requires a dedicated flow extraction pipeline (e.g., CICFlowMeter), which we leave for future work.
3. **Static λ in Sugeno integral.** The λ parameter is shared across all communities and trained globally. Per-community or per-class λ values may better capture dataset-specific co-occurrence patterns among attack families.

7 Conclusion

We presented **FC-GNN**, the first method to simultaneously provide (i) fuzzy-rule-based GNN aggregation for uncertainty-aware cybersecurity anomaly detection, (ii) distribution-free, community-conditional coverage guarantees via Fuzzy Mondrian Conformal Prediction, and (iii) human-interpretable IF-THEN rule explanations aligned with MITRE ATT&CK. Experiments on seven cybersecurity benchmarks confirm that FC-GNN is the only evaluated method to reliably maintain $\geq 90\%$ empirical coverage with conditional validity across graph communities, while achieving prediction-set efficiency competitive with or superior to the leading Mondrian CP baseline (RR-GNN) on three datasets. A formal coverage theorem establishes the finite-sample conditional coverage slack as $O(|Q_m|^{-1/2})$ under the fuzzy permutation-invariance condition.

Future work.

Three directions are most promising: (1) *Federated FC-GNN* — extending FMCP to the federated GNN setting [19, 33] where calibration data is distributed across clients without centralisation; (2) *Temporal FC-GNN* — incorporating non-exchangeable CP [21] to provide valid coverage under concept drift without retraining from scratch; (3) *Per-community Sugeno calibration* — learning community-specific λ parameters to tighten the coverage gap towards zero in all communities simultaneously.

References

- [1] Canadian Institute for Cybersecurity: CIC-IDS-2017 Dataset (2017). <https://www.unb.ca/cic/datasets/ids-2017.html>
- [2] UNSW Canberra: UNSW-NB15 Dataset (2015). <https://research.unsw.edu.au/projects/unsw-nb15-dataset>
- [3] Koroniotis, N., et al.: NF-BoT-IoT Dataset (2020). https://staff.itee.uq.edu.au/marius/NIDS_datasets/

- [4] Guerra, P.-M., Chapuis, G., Duc, A., Mozharovskyi, P., Nguyen, T.-L.: GraphIDS: Self-supervised GNN for network intrusion detection. In: Advances in Neural Information Processing Systems (NeurIPS) (2025)
- [5] Zhou, X., Chen, B., Gui, Y., Cheng, L.: Conformal prediction: A data perspective. *ACM Comput. Surv.* **58**(2), 49–14937 (2026) <https://doi.org/10.1145/3736575>
- [6] Huang, K., Jin, Y., Candès, E.J., Leskovec, J.: Uncertainty quantification over graph with conformalized graph neural networks. In: Advances in Neural Information Processing Systems (NeurIPS) (2023). arXiv:2305.14535
- [7] Zargarbashi, S.H., Antonelli, S., Bojchevski, A.: Conformal prediction sets for graph neural networks. In: Proceedings of the International Conference on Machine Learning (ICML). PMLR, vol. 202 (2023)
- [8] Clarkson, J.: Distribution free prediction sets for node classification. In: International Conference on Machine Learning, pp. 6268–6278 (2023). PMLR
- [9] Zhang, Y., Bao, J., Zhou, Z., Colombo, N., Cheng, L., Luo, W.: Residual reweighted conformal prediction for graph neural networks. In: Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI). PMLR, vol. 286, pp. 4982–4999 (2025). arXiv:2506.07854
- [10] Du, B., Zhou, J., Liu, L., She, X.: Fl-gnn: Efficient fusion of fuzzy neural network and graph neural network. In: ECAI 2024: 27th European Conference on Artificial Intelligence, 19–24 October 2024, Santiago de Compostela, Spain—including 13th Conference on Prestigious Applications of Intelligent Systems (PAIS 2024), pp. 1768–1775 (2024). SAGE Publications Pvt. Ltd 1 Oliver’s Yard, 55 City Road, London, EC1Y 1SP
- [11] Xing, J., Xing, R., Xue, C., Luo, D.: Enhancing link prediction with fuzzy graph attention networks and dynamic negative sampling. In: International Conference on Multimedia Systems and Signal Processing, pp. 187–197 (2025). Springer
- [12] Tran, V.-T., Tong, T.M., Nguyen, T.-N., Nguyen, T.-H.: Fuzzy graph neural networks: A comprehensive review of uncertainty-aware graph learning. *EAI Endorsed Transactions on Context-Aware Systems and Applications* **10** (2025)
- [13] Angelopoulos, A.N., Bates, S.: A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv preprint arXiv:2107.07511 (2021)
- [14] Barber, R.F., Candès, E.J., Tibshirani, R.J., Wager, S.: Conformal prediction beyond exchangeability. *The Annals of Statistics* **51**(2), 816–845 (2023)
- [15] Dang, Q.-V.: Conformal prediction in the intrusion detection problem. *Journal of Information Assurance and Security* **18**(1) (2023)

- [16] Dang, Q.-V., Pham, T.-H.: Kernel methods for conformal prediction to detect botnets. In: Proceedings of the International Conference on Artificial Intelligence and Its Applications (AITA). Lecture Notes in Networks and Systems, vol. 843. Springer, ??? (2024)
- [17] Dang, Q.-V., Nguyen, N.-S.-A., Vo, T.-B.-D.: CONFIDE: CONformal free inference for distribution-free estimation in causal competing risks. *Mathematics* **14**(2), 383 (2026) <https://doi.org/10.3390/math14020383>
- [18] Song, D., Huang, W., Jiang, T., Zhang, Y., Li, Y., Wang, B.: Similarity-navigated conformal prediction for graph neural networks. In: Advances in Neural Information Processing Systems (NeurIPS) (2024). arXiv:2405.14303
- [19] Akgül, A., Kannan, S., Prasanna, V.: Conformal prediction for federated GNNs with missing neighbor information. In: Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI). PMLR, vol. 286, pp. 45–63 (2025)
- [20] Davis, E., Gallagher, I., Lawson, D.J., Rubin-Delanchy, P.: Valid conformal prediction for dynamic GNNs. arXiv preprint arXiv:2405.19230 (2025)
- [21] Wang, Y., Kang, J., Yan, J., Kulkarni, S., Zhou, Z.: Non-exchangeable conformal prediction for temporal GNNs. arXiv preprint arXiv:2507.02151 (2025)
- [22] Wang, T., Zhou, Z., Luo, R.: Enhancing trustworthiness of graph neural networks with rank-based conformal training. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, pp. 21261–21268 (2025)
- [23] Bai, S., Zheng, X., Zeng, D.D.: Crc-sgad: Conformal risk control for supervised graph anomaly detection. In: 2025 IEEE International Conference on Intelligence and Security Informatics (ISI), pp. 151–156 (2025). IEEE
- [24] Koning, N., Meer, R.: Fuzzy prediction sets: Conformal prediction with E-values. arXiv preprint arXiv:2509.13130 (2026)
- [25] Xing, J., Luo, D., Cheng, Q., Xue, C., Xing, R.: Multiview fuzzy graph attention networks for enhanced graph learning. In: Tenth International Workshop on Pattern Recognition, vol. 13819, pp. 62–70 (2025). SPIE
- [26] Khaled, Q., De Marinis, P., Louati, M., Ferras, D., Genga, L., Kaymak, U.: Explainable fuzzy GNNs for leak detection in water distribution networks. arXiv preprint arXiv:2601.03062 (2026)
- [27] Sengupta, P., Rekik, I.: FireGNN: Neuro-symbolic GNNs with trainable fuzzy rules. arXiv preprint arXiv:2509.10510 (2025)
- [28] Dang, Q.-V.: Fuzzy embedding to detect intrusion in software-defined networks. In: Intelligent and Fuzzy Systems. Lecture Notes in Networks and Systems. Springer,

??? (2024). doi:10.1007/978-3-031-67195-1_78

- [29] Sun, Z., Teixeira, A.M., Toor, S.: Gnn-ids: Graph neural network based intrusion detection system. In: Proceedings of the 19th International Conference on Availability, Reliability and Security, pp. 1–12 (2024)
- [30] Farrukh, Y.A., Wali, S., Khan, I., Bastian, N.D.: XG-NID: Dual-modality network intrusion detection using a heterogeneous graph neural network and large language model. *Expert Systems with Applications* **287**, 128089 (2025)
- [31] Bilot, T., El Madhoun, N., Al Agha, K., Zouaoui, A.: Graph neural networks for intrusion detection: A survey. *IEEe Access* **11**, 49114–49139 (2023)
- [32] Dang, Q.-V., Nguyen, T.-L.: Detecting intrusion in WiFi network using graph neural networks. In: Proceedings of the International Conference on Computer, Communication, Control and Electronics Engineering (ICCCES). Springer, ??? (2023)
- [33] Dang, Q.-V., Pham, T.-H.: Detecting IoT malware using federated learning. In: Communications in Computer and Information Science. Springer, ??? (2024)
- [34] Yumlembam, R., Issac, B., Jacob, S.M., Yang, L.: Comprehensive botnet detection by mitigating adversarial attacks, navigating the subtleties of perturbation distances and fortifying predictions with conformal layers. *Information Fusion* **111**, 102529 (2024)
- [35] García, D.E., DeCastro-García, N.: Conformal prediction for labelling and updating online models in the presence of concept drift in cybersecurity. *Journal of Information Security and Applications* **93**, 104120 (2025)
- [36] Qaiser, G., Chandrasekaran, S.: Denoising adaptive multi-branch architecture for detecting cyber attacks in industrial internet of services. *Journal of Cybersecurity and Privacy* **6**(1), 26 (2026)
- [37] Yuan, S., Li, J., Wang, C., Zhang, X.: Conformal machine learning for reliable anomaly detection in industrial cyber-physical systems. *Reliability Engineering & System Safety*, 112417 (2026)
- [38] Qiu, J., Shi, L., Fan, Y.: A cooperative IDS for IoT using fuzzy logic and CNN ensemble. *Scientific Reports* **15**, 15934 (2025)
- [39] Johnpeter, T., Karuppanan, S.: Fuzzy-rule based optimized hybrid deep learning model for network intrusion detection in sdn enabled iot network. *Computers & Security* **152**, 104372 (2025) <https://doi.org/10.1016/j.cose.2025.104372>
- [40] Elhag, S., Fernández, A., Bawakid, A., Alshomrani, S., Herrera, F.: On the combination of genetic fuzzy systems and pairwise learning for improving detection

- rates on intrusion detection systems. *Expert Systems with Applications* **42**(1), 193–202 (2015)
- [41] Lal, R., Singh, R.K., Nishad, D.K., Khalid, S.: Enhancing security in electromagnetic radiation therapy using fuzzy graph theory. *Scientific Reports* **15**(1), 13139 (2025)
 - [42] Anand, M., Muthurajkumar, S.: An intelligent ids using bagging based fuzzy cnn for secured communication in vehicular networks. *Scientific Reports* **15**(1), 26952 (2025)
 - [43] Vovk, V., Gammerman, A., Shafer, G.: *Algorithmic Learning in a Random World*. Springer, ??? (2005)
 - [44] Sugeno, M.: *Theory of fuzzy integrals and its applications*. PhD thesis, Tokyo Institute of Technology (1974)
 - [45] Blondel, V.D., Guillaume, J.-L., Lambiotte, R., Lefebvre, E.: Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment* **2008**(10), 10008 (2008)
 - [46] University of New Brunswick: ISCX Botnet 2014 Dataset (2014). <https://www.unb.ca/cic/datasets/botnet.html>
 - [47] Czech Technical University: CTU-13 Botnet Dataset (2011). <https://www.stratosphereips.org/datasets-ctu13>
 - [48] Meidan, Y., et al.: N-BaIoT: Network-based Detection of IoT Botnet Attacks (2018). <https://archive.ics.uci.edu/dataset/442>
 - [49] UNSW: ToN-IoT Dataset (2020). <https://research.unsw.edu.au/projects/toniot-datasets>
 - [50] Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: *The Fifth International Conference on Learning Representations (ICLR)* (2017)
 - [51] Hamilton, W.L., Ying, R., Leskovec, J.: Inductive representation learning on large graphs. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2017)
 - [52] Zeng, H., Zhou, H., Srivastava, A., Kannan, R., Prasanna, V.: GraphSAINT: Graph sampling based inductive learning method. In: *The Eighth International Conference on Learning Representations (ICLR)* (2020)